Modeling and Analysis of a Linear Resonant Electrostatic Induction Motor Considering Capacitance Imbalance

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***Abstract*—This paper reports the modeling and analysis of a linear resonant electrostatic induction motor, with an imbalance in capacitance taken into consideration. The motor is an asynchronous motor consisting of film-type stator and slider with three-phase electrodes, driven by three-phase voltage applied to the stator. The slider electrodes are connected to three coils, which enable the slider voltage to rise high at resonance, producing large thrust force. The motor is highly sensitive to an imbalance of capacitance, particularly at resonance. A model is developed to relate each capacitance formed among the electrodes to symmetrical components of the slider voltage and to the thrust force of the motor. The model indicated that the capacitance imbalance leads to an unbalanced slider voltage with two frequency components and causes thrust force ripple. These nonideal effects were confirmed by experiments, using a motor with geometric asymmetry in its electrode design.**

***Index Terms*—Actuators, electrostatic actuators, electrostatic induction, resonance.**

NOMENCLATURE

a	Unit complex operator $= \exp(j2\pi/3)$.
C_{jk}	Capacitance between phases j and k (in farads).
$\bar{C}_{jk}$	Offset of gap capacitance (in farads).
C_k	Self-capacitance of phase k (in farads).
C_m	Amplitude of gap capacitance (in farads).
F	Thrust force (in newtons).
i	Temporal current (in amperes).
$\dot{I}$	Time-independent current phasor (in $\mathrm{A_{rms}}$).
K_F	Force constant $(= 9\kappa C_m/2)$ (in $\mathrm{N/V_{rms}^2}$).
L	Inductance of a coil (in henrys).
p	Electrode pitch (in meters).
R	Resistance of a coil (in ohms).
u	Slider speed $(= dx/dt)$ (in meters per second).
v	Temporal voltage (in volts).
V	Magnitude of the voltage phasor (in $\mathrm{V_{rms}}$).
$\dot{V}$	Time-independent voltage phasor (in $\mathrm{V_{rms}}$).
V_t	RMS voltage of excitation (in $\mathrm{V_{rms}}$).
x	Slider position (in meters).
κ	Angular wavenumber $(= 2\pi/3p)$ (in radians per meter).
ϕ	Phase angle of the voltage phasor (in radians).
ω	Angular frequency of excitation (in radians per second).
ω_s	Angular slip frequency $(= \omega - \kappa u)$ (in radians per second).

SUBSCRIPTS

1, 2, 3	Phase numbers given to stator electrodes.
4, 5, 6	Phase numbers given to slider electrodes.
s	Slip frequency component.
e	Excitation frequency component.
$+$	Positive-sequence component.
$-$	Negative-sequence component.

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I. INTRODUCTION

A RESONANT electrostatic induction motor (REIM) is an asynchronous electrostatic motor that has three-phase electrodes patterned on both of its stator and rotor, with electrodes in the latter connected to coils [1]. The coils enable the motor to produce large torque at resonance, where voltages on the rotor rise up high. The motor can take various forms: It can be either rotary or linear [2], and the coils can be either placed outside or mounted on the motor. The rotary motor could exert 20 mN · m of torque [1], and the linear motor could pull up 80 g of weight [2]; the torque/force is considerably large for an electrostatic motor. The motor has an advantage of being thin: The stator and the rotor/slider are flexible printed circuits, which are derived from electrostatic film motors [3], [4]. With such advantage, the motor is expected to be utilized in special mechatronic systems in which space efficiency is the primary requirement for their actuators.

There are two optimization problems in designing the REIM. One is optimization of coils, particularly in downsizing; the rotor/slider loses its thinness with mounted coils, which leaves the stator the only thin component. An ongoing attempt for the downsizing is replacing the coils with piezoelectric elements operating at frequencies in which they behave inductively [5]. Another optimization problem is in the design of electrodes. Such necessity comes from the recent finding that the motor is highly sensitive to an imbalance of capacitance; the imbalance causes voltage imbalance and torque/force ripple [6]. This paper reports an analysis of a linear REIM with such imbalance, focusing on the effect of the imbalance on the motor's thrust force.

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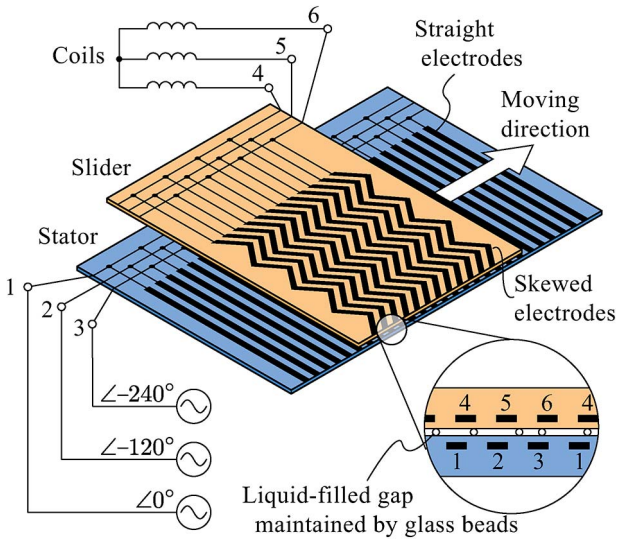


Fig. 1. Schematic of a linear REIM.

The problem of capacitance imbalance is unique to the REIM among electrostatic induction motors, and the uniqueness comes from its rotor structure with three-phase electrodes. Most electrostatic induction motors studied in the past had either bulk dielectric rotors with smooth surface or those with thin conductive layers [7]–[12]. Motors with such rotor structure have an advantage of being simple, with which micromotors and generators with promising performance were realized in the field of microelectromechanical systems [13]–[15]. Some motors had rotors in other forms: The motor proposed by Charpentier *et al.* had salient poles [16], and dielectric motors of many different rotor shapes were developed and tested by Hagedorn *et al.* [17]. However, none of these motors had polyphase electrodes in their rotor; thus, analysis concerning the capacitance imbalance was not necessary.

The remainder of this paper is as follows. Section II briefly describes the basic configuration of the motor and its ideal characteristic. The section also provides an example of electrode pattern that causes the capacitance imbalance. In Section III, a model of the REIM with capacitance imbalance is constructed and analyzed. The analysis bases on the method of symmetrical components, which is a well-established method for asymmetry analysis of polyphase power systems found by Fortesucue in 1918 [18]. The result of the analysis shows that the imbalance not only causes an imbalance of slider voltages but also causes the voltages to have multiple frequency components, which give rise to a thrust force ripple. Section IV compares the analytical results with experimental results. The last two sections are discussion and conclusion, respectively.

II. Structure and Principle of REIM

A. Structure of REIM

Fig. 1 shows a schematic of a linear REIM. The motor is composed of a pair of films and three coils. The two films, respectively referred to as the stator film and the slider film, are flexible printed circuits, with three-phase electrodes made of copper embedded in their polyimide substrate. The slider

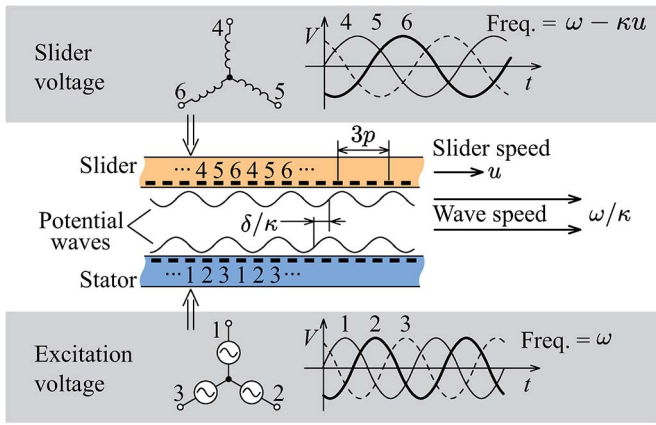


Fig. 2. Relation between traveling potential waves and voltages on electrodes, in an ideal REIM.

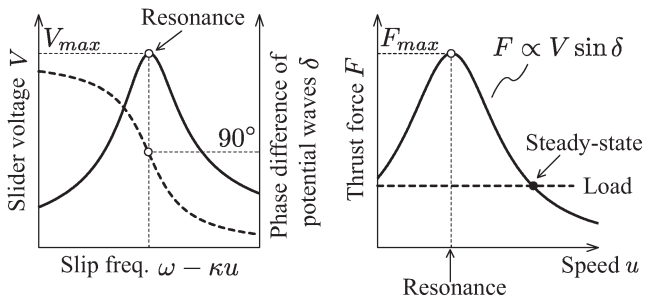


Fig. 3. Ideal force-slip characteristic of REIM.

electrodes are skewed, as proposed in [19], to reduce force ripples. The stator electrodes are connected to a three-phase voltage source, and the slider electrodes are connected to three coils in wye configuration. In the gap between the films are glass beads and dielectric liquid; the former serve as spacers and lubricators, and the latter enables the electric field in the gap to exceed the breakdown strength of air.

B. Principle and Ideal Characteristic

As illustrated in Fig. 2, voltages on the three-phase electrodes form traveling potential waves with angular wavenumber $\kappa = 2\pi/3p$ on the stator and the slider. The stator wave travels at a speed of ω/κ, where ω is the angular frequency of the source. The voltages on the slider electrodes are passively determined, depending on the excitation voltage and the slider speed u. The slider voltages alternate at a frequency called the slip frequency: $\omega_s = \omega - \kappa u$, forming a potential wave traveling at the speed $\omega/\kappa - u$ with respect to the slider. Therefore, seen from the stator, the two potential waves travel in synchrony at the speed ω/κ. These traveling waves form a traveling electric field in the gap, and the tangential component of the field gives thrust to the slider.

Fig. 3 shows the ideal characteristic of the REIM. When the slip frequency is equal to the resonance frequency, amplitudes of the slider voltages reach their peak value, and the spatial phase difference of the waves (δ in Fig. 2) approaches 90°. At this condition, the motor produces the maximum thrust force. At steady-state condition with a constant load, the motor travels at the speed where the thrust force balances with the load.

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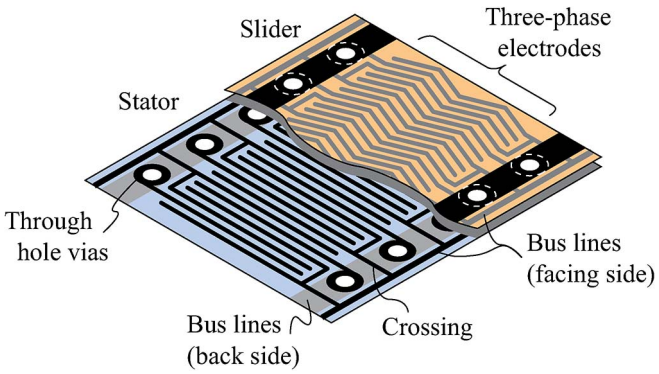


Fig. 4. Electrode pattern with asymmetry in bus lines and through holes.

C. Capacitance Imbalance from Geometric Asymmetry

The aforementioned explanations assume that the slider voltage is a balanced three-phase voltage in the same phase sequence as the stator side. This comes from an idealization that the capacitances among the electrodes are perfectly symmetric. However, such idealization is invalid when geometric asymmetry exists in electrodes. Fig. 4 is a schematic of the electrodes used in the experiments of this work. In this design, the three-phase electrodes are symmetric, but the bus lines and through holes are asymmetric. Some pairs of bus lines are facing each other across the gap, and some lines are crossing the bus lines of other phases. The former is avoidable by separating all bus lines by distance, but the latter is unavoidable because the crossing of lines is always necessary in three-phase electrodes. Capacitances formed among such asymmetric conductors cause an imbalance of slider voltage and a ripple of thrust force, as reported in [6].

III. Modeling of REIM With Capacitance Imbalance

This section describes the model of an REIM with the capacitance imbalance taken into consideration. The goal of the analysis is to relate the imbalance to the thrust force. A large part of the analysis is concerning the determination of the slider voltages, which is necessary to calculate the force.

A. Basic Assumptions and Differential Equations

Fig. 5 shows the circuit model of the REIM, based on the capacitance network model proposed in [20]. Assuming optimally skewed electrodes, gap capacitances, the capacitances connecting stator and slider nodes, can be approximated as sinusoidal functions of slider position x as

$$C_{jk} = \bar{C}_{jk} + C_m \cos(\kappa x + \varphi_{jk})$$
$$\varphi_{jk} = \begin{cases} 0, & (j,k) = (1,4),(2,5), \text{ and } (3,6) \\ 2\pi/3, & (j,k) = (1,5),(2,6), \text{ and } (3,4) \\ -2\pi/3, & (j,k) = (1,6),(2,4), \text{ and } (3,5). \end{cases} \quad (1)$$

Slider capacitances and stator capacitances that connect two nodes in the same film are independent of position. The idealized model in [1] assumed symmetry of capacitances, i.e.,

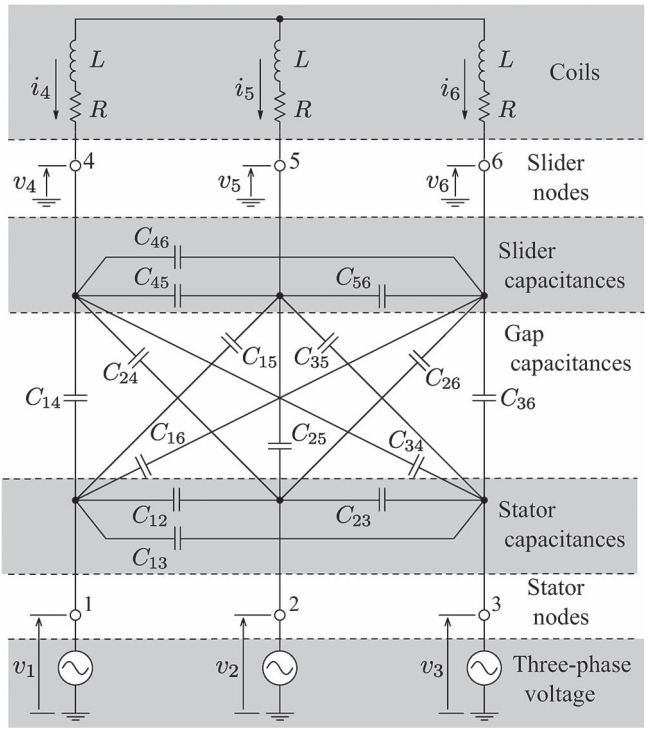


Fig. 5. Circuit model of REIM.

$C_{45} = C_{46} = C_{56}$ and all of the capacitance offsets $\bar{C}_{jk}$ being equal. To include capacitance imbalance, the present model assumes neither of these.

As an excitation voltage to the stator, we assume a balanced three-phase voltage as

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \sqrt{2} V_t \begin{bmatrix} \sin(\omega t) \\ \sin(\omega t - 2\pi/3) \\ \sin(\omega t + 2\pi/3) \end{bmatrix}. \quad (2)$$

After performing trigonometric simplifications, the slider currents are related to the voltages as

$$\begin{bmatrix} i_4 \\ i_5 \\ i_6 \end{bmatrix} = -\frac{3}{2} C_m \frac{d}{dt} \begin{bmatrix} v_{s1} \\ v_{s2} \\ v_{s3} \end{bmatrix} - \boldsymbol{C}_0 \frac{d}{dt} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \boldsymbol{C}_l \frac{d}{dt} \begin{bmatrix} v_4 \\ v_5 \\ v_6 \end{bmatrix}. \quad (3)$$

The symbols $\boldsymbol{C}_0$ and $\boldsymbol{C}_l$ are three-by-three matrices respectively defined as

$$\boldsymbol{C}_0 = \begin{bmatrix} \bar{C}_{14} & \bar{C}_{15} & \bar{C}_{16} \\ \bar{C}_{24} & \bar{C}_{25} & \bar{C}_{26} \\ \bar{C}_{34} & \bar{C}_{35} & \bar{C}_{36} \end{bmatrix} \qquad \boldsymbol{C}_l = \begin{bmatrix} C_4 & -C_{45} & -C_{46} \\ & C_5 & -C_{56} \\ & \text{sym.} & C_6 \end{bmatrix} \quad (4)$$

and voltages v_{s1}, v_{s2}, and v_{s3} are defined as

$$\begin{bmatrix} v_{s1} \\ v_{s2} \\ v_{s3} \end{bmatrix} = \sqrt{2} V_t \begin{bmatrix} \sin(\omega t - \kappa x) \\ \sin(\omega t - \kappa x - 2\pi/3) \\ \sin(\omega t - \kappa x + 2\pi/3) \end{bmatrix}. \quad (5)$$

Diagonal elements of $\boldsymbol{C}_l$ are the self-capacitances of slider electrodes, defined as the sum of all capacitances connected to a node, e.g.,

$$C_4 = \bar{C}_{14} + \bar{C}_{24} + \bar{C}_{34} + C_{45} + C_{46}. \quad (6)$$

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The self-capacitances are constants, because position-dependent terms cancel out in summation.

Differential equation regarding the coils is simply obtained. Eliminating the voltage on the neutral node of wye using Kirchoff's current law $i_4 + i_5 + i_6 = 0$, the equation is

$$\frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}\begin{bmatrix} v_4 \\ v_5 \\ v_6 \end{bmatrix} = -R\begin{bmatrix} i_4 \\ i_5 \\ i_6 \end{bmatrix} - L\frac{d}{dt}\begin{bmatrix} i_4 \\ i_5 \\ i_6 \end{bmatrix}. \tag{7}$$

B. Phasor Equations

To solve (3) and (7) for the slider voltages, the time invariance of the slider speed $u = dx/dt$ is assumed. At such condition, voltages v_{s1}, v_{s2}, and v_{s3} in (8) have constant frequency $\omega_s = \omega - \kappa u$, which is the slip frequency. In steady state, the slider position is $x = ut + x_0$, where x_0 is the initial position at $t = 0$. With the steady-state assumption, voltages in (5) are rewritten as

$$\begin{bmatrix} v_{s1} \\ v_{s2} \\ v_{s3} \end{bmatrix} = \sqrt{2}V_t \begin{bmatrix} \sin(\omega_s t - \kappa x_0) \\ \sin(\omega_s t - \kappa x_0 - 2\pi/3) \\ \sin(\omega_s t - \kappa x_0 + 2\pi/3) \end{bmatrix}. \tag{8}$$

Thus, (3) has two forcing terms with different frequencies. With these forcing terms, the slider voltages and currents are expected to have the forms

$$\begin{bmatrix} v_4 \\ v_5 \\ v_6 \end{bmatrix} = \sqrt{2}\mathrm{Im}\left(\begin{bmatrix} \dot{V}_{s4} \\ \dot{V}_{s5} \\ \dot{V}_{s6} \end{bmatrix} \exp j(\omega_s t - \kappa x_0) + \begin{bmatrix} \dot{V}_{e4} \\ \dot{V}_{e5} \\ \dot{V}_{e6} \end{bmatrix} \exp j\omega t \right)$$
$$\begin{bmatrix} i_4 \\ i_5 \\ i_6 \end{bmatrix} = \sqrt{2}\mathrm{Im}\left(\begin{bmatrix} \dot{I}_{s4} \\ \dot{I}_{s5} \\ \dot{I}_{s6} \end{bmatrix} \exp j(\omega_s t - \kappa x_0) + \begin{bmatrix} \dot{I}_{e4} \\ \dot{I}_{e5} \\ \dot{I}_{e6} \end{bmatrix} \exp j\omega t \right). \tag{9}$$

Sets of phasor equations for each frequency are obtained by substituting (2), (8), and (9) into (7) and (3) and by comparing the terms with the same frequencies. The phasor equations for the slip frequency components are

$$\begin{cases} \begin{bmatrix} \dot{I}_{s4} \\ \dot{I}_{s5} \\ \dot{I}_{s6} \end{bmatrix} = -\frac{3}{2}j\omega_s C_m V_t \begin{bmatrix} 1 \\ a^2 \\ a \end{bmatrix} + j\omega_s \boldsymbol{C}_l \begin{bmatrix} \dot{V}_{s4} \\ \dot{V}_{s5} \\ \dot{V}_{s6} \end{bmatrix} \\ \begin{bmatrix} \dot{I}_{s4} \\ \dot{I}_{s5} \\ \dot{I}_{s6} \end{bmatrix} = -\frac{1}{3}(R + j\omega_s L)^{-1} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \dot{V}_{s4} \\ \dot{V}_{s5} \\ \dot{V}_{s6} \end{bmatrix} \end{cases} \tag{10}$$

and those for the excitation frequency components are

$$\begin{cases} \begin{bmatrix} \dot{I}_{e4} \\ \dot{I}_{e5} \\ \dot{I}_{e6} \end{bmatrix} = -j\omega \boldsymbol{C}_0 V_t \begin{bmatrix} 1 \\ a^2 \\ a \end{bmatrix} + j\omega \boldsymbol{C}_l \begin{bmatrix} \dot{V}_{e4} \\ \dot{V}_{e5} \\ \dot{V}_{e6} \end{bmatrix} \\ \begin{bmatrix} \dot{I}_{e4} \\ \dot{I}_{e5} \\ \dot{I}_{e6} \end{bmatrix} = -\frac{1}{3}(R + j\omega L)^{-1} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} \dot{V}_{e4} \\ \dot{V}_{e5} \\ \dot{V}_{e6} \end{bmatrix}. \end{cases} \tag{11}$$

C. Symmetrical Component Transformation

It is well known that any set of unbalanced three-phase phasors can be expressed as a sum of three symmetrical components: zero-, positive-, and negative-sequence components. In this paper, zero-sequence components are neglected for simplicity. The slider voltages are related to the rest of the components as

$$\begin{bmatrix} \dot{V}_{k4} \\ \dot{V}_{k5} \\ \dot{V}_{k6} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ a^2 & a \\ a & a^2 \end{bmatrix} \begin{bmatrix} \dot{V}_{k+} \\ \dot{V}_{k-} \end{bmatrix} \qquad (k = s, e). \tag{12}$$

As for the current phasors, they are related to the positive- and positive-sequence components as

$$\begin{bmatrix} \dot{I}_{k+} \\ \dot{I}_{k-} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \dot{I}_{k4} \\ \dot{I}_{k5} \\ \dot{I}_{k6} \end{bmatrix} \qquad (k = s, e). \tag{13}$$

From (10)–(13), solutions for each component of the slider voltages are obtained in simple forms. For the slip frequency components, the expressions are

$$\begin{cases} \dot{V}_{s+} = \dot{V}_{sL} + \dot{V}_{sH} \\ \dot{V}_{s-} = (\dot{V}_{sL} - \dot{V}_{sH}) \exp(j\theta) \end{cases} \tag{14}$$

where

$$\dot{V}_{sL}, \dot{V}_{sH} = \frac{3}{4}\frac{j\omega_s C_m}{(R + j\omega_s L)^{-1} + j\omega_s (C \pm \Delta C)} V_t. \tag{15}$$

The capacitance C is defined as

$$C = (C_4 + C_5 + C_6 + C_{45} + C_{46} + C_{56})/3 \tag{16}$$

and the capacitance ΔC and the phase angle θ are defined as

$$\Delta C \exp(j\theta) = \frac{1}{3}\left\{C_4 - 2C_{56} + a(C_5 - 2C_{46}) + a^2(C_6 - 2C_{45})\right\}. \tag{17}$$

The voltage phasors of the excitation frequency components are expressed as

$$\begin{cases} \dot{V}_{e+} = \dot{V}_{eL} + \dot{V}_{eH} \\ \dot{V}_{e-} = (\dot{V}_{eL} - \dot{V}_{eH}) \exp(j\theta) \end{cases} \tag{18}$$

where

$$\dot{V}_{eL}, \dot{V}_{eH} = \frac{1}{2}\frac{j\omega \left\{C_a \exp(j\alpha) \pm C_b \exp(j\beta - j\theta)\right\}}{(R + j\omega L)^{-1} + j\omega(C \pm \Delta C)} V_t. \tag{19}$$

Capacitances C_a and C_b and phase angles α and β are calculated from

$$C_a \exp(j\alpha) = \frac{1}{3}\left\{\bar{C}_{14} + \bar{C}_{25} + \bar{C}_{36} + a(\bar{C}_{15} + \bar{C}_{26} + \bar{C}_{34}) + a^2(\bar{C}_{16} + \bar{C}_{24} + \bar{C}_{35})\right\}$$
$$C_b \exp(j\beta) = \frac{1}{3}\left\{\bar{C}_{14} + \bar{C}_{26} + \bar{C}_{35} + a(\bar{C}_{16} + \bar{C}_{25} + \bar{C}_{34}) + a^2(\bar{C}_{15} + \bar{C}_{24} + \bar{C}_{36})\right\}. \tag{20}$$

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In a perfectly balanced motor, in which assumptions $C_4 = C_5 = C_6$ and $C_{45} = C_{46} = C_{56}$ and all elements of $\boldsymbol{C_0}$ being equal are valid, $\Delta C = C_a = C_b = 0$. Thus, all of the nonideal voltage components ($\dot{V}_{s-}$, $\dot{V}_{e+}$, and $\dot{V}_{e-}$) become null. The only component left is $\dot{V}_{s+}$, the positive-sequence voltage of the slip frequency, and it becomes

$$\dot{V}_{s+,\text{balanced}} = \frac{3}{2}\frac{j\omega_s C_m}{(R+j\omega_s L)^{-1}+j\omega_s C}\,V_t \tag{21}$$

because $\dot{V}_{sL} = \dot{V}_{sH}$. This has the same form as the idealized model in [1].

When there is an imbalance in the slider capacitances or self-capacitances, $\Delta C \neq 0$. Thus, the positive-sequence component $\dot{V}_{s-}$ emerges. Furthermore, if there is an imbalance in the offsets of gap capacitances, C_a and C_b are not null. Hence, excitation frequency components become present.

D. Slider Voltage in Temporal Form

The phasors of the two frequency components were introduced for the convenience of the analysis, but they cannot be measured separately in an actual motor. Thus, it is important to express the slider voltages in temporal forms and examine how these components appear in the voltage waveforms. From (9) and (12), the slider voltages in temporal forms are

$$\begin{bmatrix} v_4 \\ v_5 \\ v_6 \end{bmatrix} = \sqrt{2}\,\{V_{s+}\boldsymbol{s}_+(\omega_s t - \kappa x_0 + \phi_{s+}) + V_{e+}\boldsymbol{s}_+(\omega t + \phi_{e+}) + V_{s-}\boldsymbol{s}_-(\omega_s t - \kappa x_0 + \phi_{s-}) + V_{e-}\boldsymbol{s}_-(\omega t + \phi_{e-})\} \tag{22}$$

where

$$\boldsymbol{s}_+(\xi) = \begin{bmatrix} \sin\xi \\ \sin(\xi - 2\pi/3) \\ \sin(\xi + 2\pi/3) \end{bmatrix} \qquad \boldsymbol{s}_-(\xi) = \begin{bmatrix} \sin\xi \\ \sin(\xi + 2\pi/3) \\ \sin(\xi - 2\pi/3) \end{bmatrix}. \tag{23}$$

The difference of the two frequencies ω_s and ω is κu; thus, in temporal waveforms, a beat pattern should appear with the frequency of their envelopes proportional to the slider speed.

In a special case when the slider is not moving, i.e., $u = 0$, the two frequencies become equal; thus, the beat patterns of voltages are not observed. Instead, the slider voltages appear to be position dependent. Assuming that $\omega_s = \omega$ in (22), terms with the same phase sequences are combined as

$$\begin{bmatrix} v_4 \\ v_5 \\ v_6 \end{bmatrix} = \sqrt{2}\,\{V_+\boldsymbol{s}_+(\omega t + \phi_+) + V_-\boldsymbol{s}_-(\omega t + \phi_-)\} \tag{24}$$

where voltages V_+ and V_- and phase angles ϕ_+ and ϕ_- are defined as

$$V_k = \sqrt{V_{sk}^2 + V_{ek}^2 + 2V_{sk}V_{ek}\cos(\kappa x_0 - \phi_{sk} + \phi_{ek})}$$
$$\phi_k = \tan^{-1}\frac{V_{sk}\sin(\phi_{sk} - \kappa x_0) + V_{ek}\sin\phi_{ek}}{V_{sk}\cos(\phi_{sk} - \kappa x_0) + V_{ek}\cos\phi_{ek}}, \quad (k = +, -). \tag{25}$$

Thus, the positive- and positive-sequence components of the slider voltage change depending on the position x_0, where the slider is standing still.

E. Forces by Symmetrical Components

The thrust force F of the motor is calculated from the virtual work principle as

$$F = \sum \frac{1}{2}\frac{\partial C_{jk}}{\partial x}\,(v_j - v_k)^2 \tag{26}$$

where the summation is done for all pairs of (j, k). Substituting (22) into (26), the thrust force is decomposed into four components

$$F = F_{s+} + F_{s-} + F_{e+} + F_{e-} \tag{27}$$

where

$$\begin{aligned} F_{s+} &= K_F V_t V_{s+}\sin\phi_{s+} \\ F_{s-} &= K_F V_t V_{s-}\sin(2\omega t - 2\kappa x + \phi_{s-}) \\ F_{e+} &= K_F V_t V_{e+}\sin(\kappa x + \phi_{e+}) \\ F_{e-} &= K_F V_t V_{e-}\sin(2\omega t - \kappa x + \phi_{e-}\}. \end{aligned} \tag{28}$$

The force by the positive-sequence voltage of slip frequency (F_{s+}) is the position-independent thrust force, which is the only force expected in an ideal REIM. The other three force components are the nonideal forces resulting from capacitance imbalance. The force by the positive-sequence voltage of excitation frequency (F_{e+}) is a position-dependent force ripple: The force changes periodically as the slider travels, with a spatial period of $3p$. Other two forces by positive-sequence voltages (F_{s-} and F_{e-}) are oscillating forces. The frequencies of these oscillations are on the order of kilohertz, much faster than the mechanical response of the REIM. Thus, such oscillations are not observed in an actual motor.

These results indicate that only the positive-sequence components of the slider voltage produce effective forces, with that of the slip frequency contributing to the thrust force and that of the excitation frequency causing force ripple.

IV. Experimental Results

This section describes the results of experiments using the linear REIM shown in Fig. 6. The experiments include voltage measurements at standstill, quasi-static force measurements, and steady-state voltage measurements.

A. Specifications of Components

Fig. 7 shows a photograph of the stator film and the slider film installed in the motor. Capacitance parameters of these films were identified by the method in [20], as shown in Table I, which indicate that imbalances are present in each class of capacitances. From these capacitances, the parameters described in the previous section are calculated as

$$\begin{cases} C = 2080\ \text{pF}, \quad \Delta C = 42\ \text{pF}, \quad \theta = 44^\circ \\ C_a = 19\ \text{pF}, \quad \alpha = 129^\circ, \quad C_b = 3.3\ \text{pF}, \quad \beta = 0^\circ \\ K_F = 4.7\ \text{N/kV}_{\text{rms}}^2. \end{cases} \tag{29}$$

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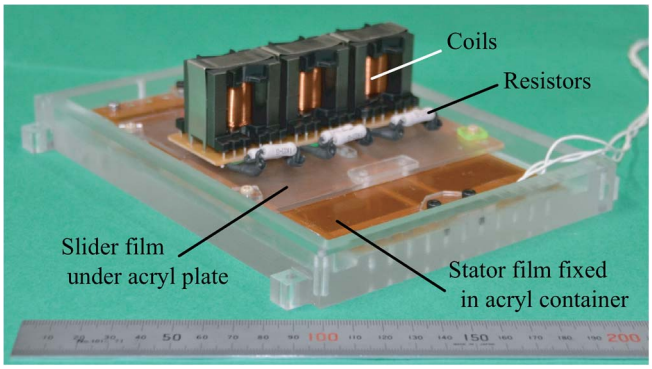


Fig. 6. Photograph of the REIM. The stator container is filled with dielectric liquid (Fluorinert FC-77, 3M), and glass beads of 20-μm diameter are scattered between the stator and slider films.

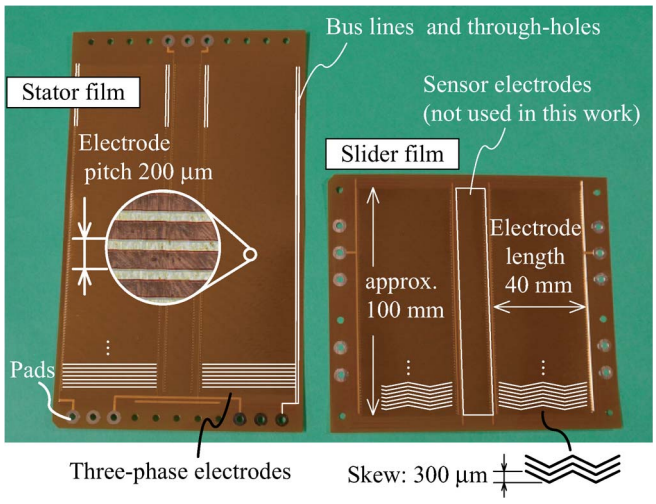


Fig. 7. Photograph of stator and slider films. Slider electrodes are skewed to reduce thrust force ripple.

TABLE I
CAPACITANCE PARAMETERS IDENTIFIED USING AN IMPEDANCE METER (ZM2372, NF CORPORATION) AT 1 V_{rms}/3 kHz

	Symbols	Capacitances (pF)
Self-capacitances	C_4 / C_5 / C_6	1590 / 1570 / 1530
Slider capacitances	C_{45} / C_{46} / C_{56}	540 / 510 / 500
Offsets, gap capacitances	$\bar{C}_{14}$ / $\bar{C}_{15}$ / $\bar{C}_{16}$ $\bar{C}_{24}$ / $\bar{C}_{25}$ / $\bar{C}_{26}$ $\bar{C}_{34}$ / $\bar{C}_{35}$ / $\bar{C}_{36}$	170 / 190 / 170 170 / 170 / 190 190 / 180 / 170
Amplitude, gap capacitances	C_m	100 pF

The three coils were wound on a ferrite core (PQ3230, TDK), and the number of turns was 1020 each. The inductances of the coils were $L = 1.28$ H, and the resistances were $R = 800\ \Omega$, identified by measuring currents while directly applying a high voltage of 700 V_{rms}, 3 kHz. In series with these coils, extra resistors of 500 Ω were added to suppress the slider voltages at resonance.

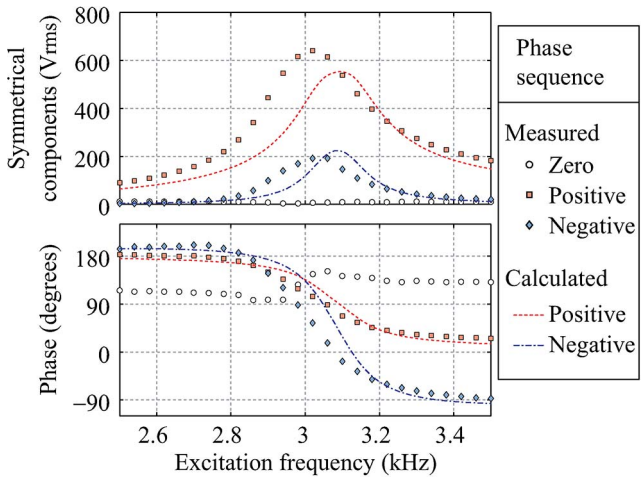


Fig. 8. Measured and calculated symmetrical components of the slider voltages with slider standing still at position $x = 0$, with an excitation voltage of 500 V_{rms}.

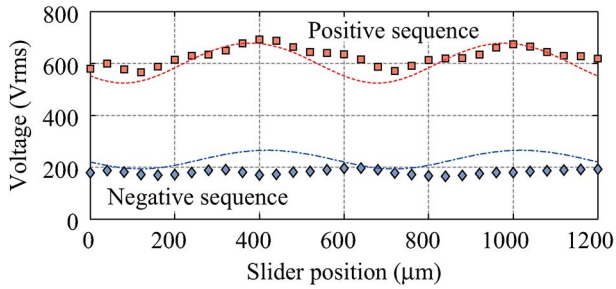


Fig. 9. Measured and calculated symmetrical components of the slider voltage with the slider standing still at different positions. The excitation voltage was fixed at 500 V_{rms}, 3.02 kHz. The lines show the calculated values using identified parameters, at 500 V_{rms}, 3.1 kHz.

B. Voltage Measurements at Standstill

The slider voltages were measured with the slider standing still, with different excitation frequencies. The procedure of measurement was as follows. First, the voltages on each of the slider electrodes were measured together with the excitation voltage applied to the stator phase 1. Then, measured voltages were expressed in phasors, with the excitation voltage as the phase reference. Finally, the voltages were decomposed into symmetrical components. The result of the measurement is shown in Fig. 8. Theoretical curves in the figure are based on (24). The results show that the calculated and measured values agree qualitatively, both showing that the slider voltage contains positive-sequence components with a magnitude approximately one third of the positive-sequence component.

The same measurements were conducted with different slider positions, and the results are shown in Fig. 9. In the measurement, the excitation frequency was fixed at 3.02 kHz, where the voltage amplitude was maximized at $x = 0$. The frequency for the theoretical values was fixed at 3.1 kHz, which is also the frequency that gives the maximum voltage. A position dependence was apparent in the measured positive-sequence voltage. As for the positive-sequence component, the position dependence was unclear.

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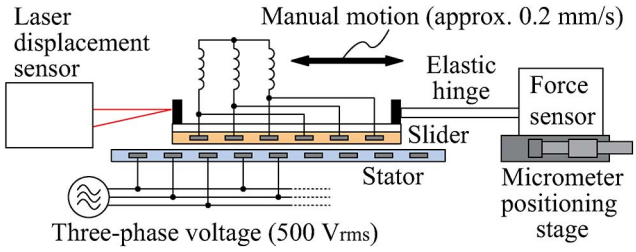


Fig. 10. Experimental setup for quasi-static force measurement.

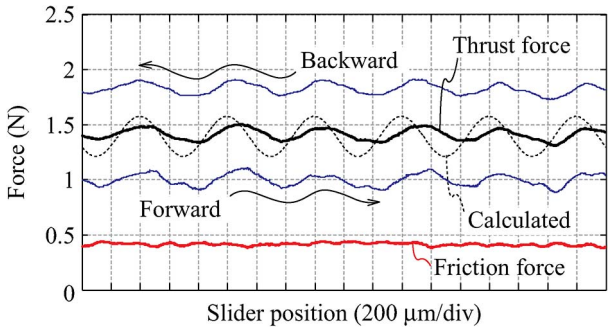


Fig. 11. Example of forces in quasi-static measurement. Excitation voltage was set at 500 V_{rms}, 3.02 kHz in the experiment. The calculated force assumes 500 V_{rms}, 3.1 kHz.

C. Quasi-Static Force Measurements

The force was measured quasi-statically, with the slider traveling at low speed, using the setup shown in Fig. 10. The procedure of the measurement was as follows. First, one end of a push–pull-type force sensor (LTS-1 KA, Kyowa Electronic Instruments) was connected to the slider through a flexure hinge. The other end of the sensor was mounted on top of a positioning stage with micrometer. Using the micrometer, the slider and the force sensor were slowly moved back and forth at approximately 0.2 mm/s, with an excitation voltage of 500 V_{rms} applied to the stator. The speed corresponds to a change of 0.33 Hz in the slip frequency, which is only 0.01% of the excitation frequency; thus, the effect of slider speed was negligible in this experiment.

An example of the measured force is shown in Fig. 11. When the slider is pulled against the thrust force of the motor, the sensor measures the sum of the thrust force and the friction between the films. On the other hand, when the motor is pushed in the same direction as the thrust force, the measured force is a difference of the thrust force and the friction. Thus, the thrust force and the friction force can be calculated from the forces in forward and backward paths.

Fig. 12 shows the average thrust force, force ripple in peak to peak, and the friction force measured at different excitation frequencies. Forces F_{s+} and F_{e+} in (28) are calculated to draw theoretical curves, respectively for the average thrust force and the force ripple. The trend of the average thrust force and its magnitude agree with the theoretical prediction. However, despite expectation, the force ripple was almost constant regardless of the frequency. This comes from the space harmonics of the force ripple shown in Fig. 13, which we excluded from our model.

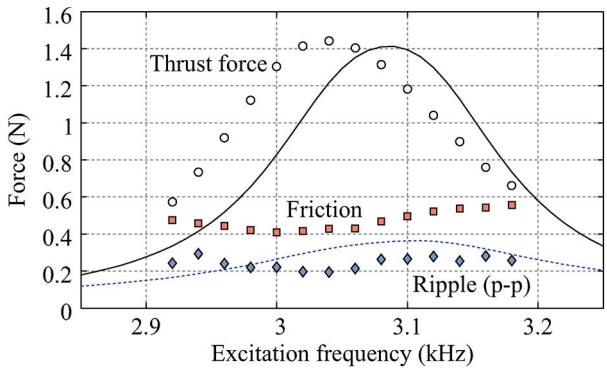


Fig. 12. Quasi-statically measured average thrust force, force ripple, and friction force of REIM, with 500-V_{rms} excitation voltage. The solid and dotted lines indicate the calculated thrust force and force ripple, respectively.

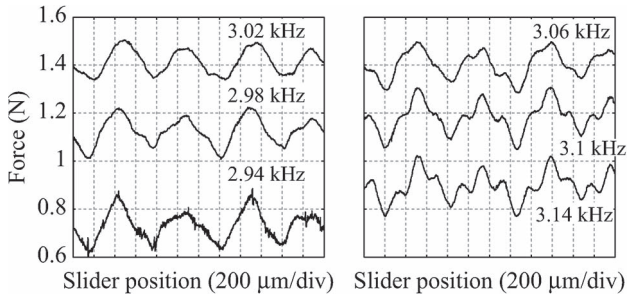


Fig. 13. Waveforms of force ripple observed in quasi-static force measurements with 500-V_{rms} excitation.

D. Steady-State Measurements

Finally, the slider voltage was measured in steady-state condition. Fig. 14 shows an example of the data, measured in the following procedure. First, the slider was fixed still, and the excitation voltage was applied to the stator. Then, the slider was set free to accelerate. After a transient acceleration, the slider reached its steady-state speed, at which the force was in balance with the friction. During the process, the slider position was recorded using a laser displacement sensor (LB-300, Keyence) simultaneously with the slider voltages.

Beat patterns are observed from the envelopes of the voltage waveforms, which indicate that the waves have multiple frequency components. Fig. 15 shows the first and second dominant frequencies of the voltages in the fast Fourier transform at steady-state condition, with various excitation frequencies. The first dominant peak appeared at the slip frequency, and the second one appeared at the excitation frequency. At all conditions, the slider reached its steady-state speed when the slip frequency was approximately 2.9 kHz; as shown in Fig. 12, the thrust force falls below the friction force at this frequency.

V. Discussion

The main contribution of this work is in providing a model of the REIM, which gives future researchers guidelines for designing electrodes. The model indicates the importance in considering the following two points regarding capacitance imbalance.

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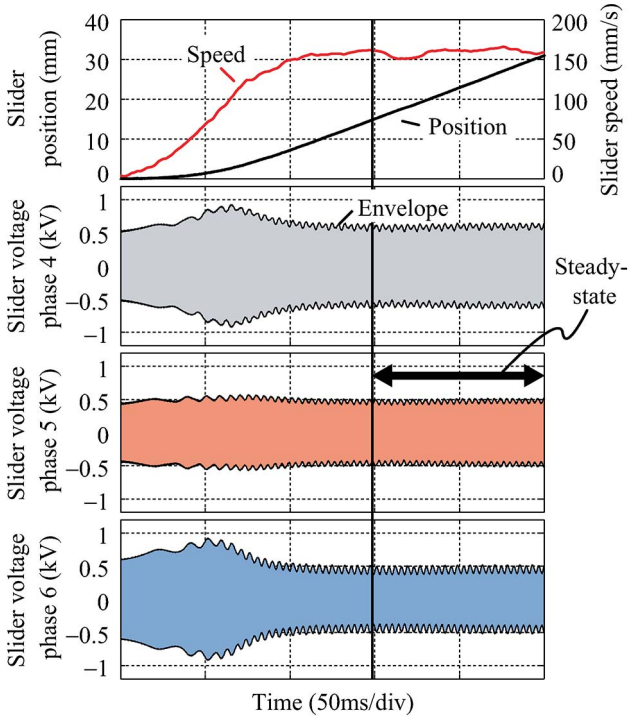


Fig. 14. Slider voltage waveforms in no-load acceleration from standstill to steady state with an excitation voltage of 500 V_{rms}, 3.18 kHz.

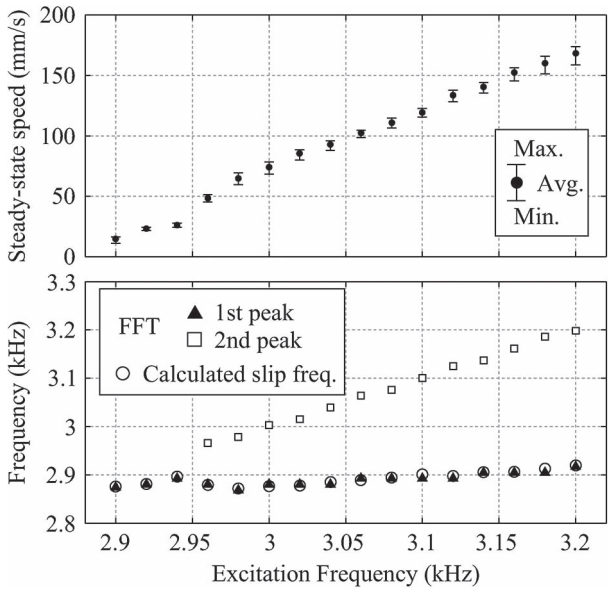


Fig. 15. Steady-state speed and dominant frequencies of slider voltage with different excitation frequencies. Slip frequencies calculated from excitation frequency and the average speed are also shown. The two dominant frequencies were indistinguishable below 2.94 kHz.

First, an imbalance in capacitances across the gap should be avoided by all means, because such imbalance causes excitation frequency components to appear in the slider voltage, which produces a thrust force ripple. For the motor used in this work, the imbalance was approximately 10% among the elements of $\boldsymbol{C}_0$ in (4), and the resulting ripple was 14% of the maximum thrust force (Fig. 12). As demonstrated in the preceding work

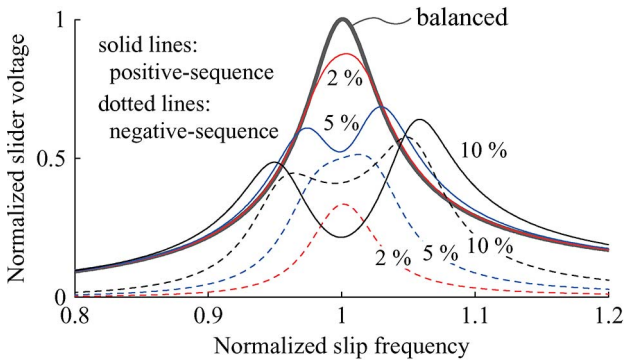


Fig. 16. Effect of capacitance imbalance on slip frequency components of slider voltage. Numbers in percentage represent the imbalance $\Delta C/C$. Voltage and slip frequency are normalized by the maximum voltage and the resonance frequency of the balanced case, respectively.

[6], this type of imbalance is easily prevented by avoiding unnecessary facing of electrode patterns across the gap; only facing patterns should be the three-phase strip electrodes. In case that practical limitations disallow this solution, one should refer to (18)–(20) to adjust the imbalance to minimize the voltage component $\dot{V}_{e+}$.

Second, to make full use of the increased voltage at resonance, slider electrodes should be as symmetric as possible, so that ΔC in (17) becomes much smaller than C in (16). Our recommendation is to keep the imbalance $\Delta C/C$ on the order of a tenth of percent, so that the positive-sequence component of the slider voltage becomes negligibly small. Fig. 16 shows the effect of the capacitance imbalance on the slip frequency component of the slider voltage. An increase in the imbalance splits the resonance peak. At each split peaks, the positive-sequence voltage is decreased, and the negative-sequence voltage is increased compared to the balanced case. The reduced positive-sequence voltage brings reduced thrust force; for example, if the present motor had 10% imbalance, the force is reduced almost by half. Even with such large imbalance, the motor still works with reduced force near the two resonance peaks. However, if the design goal is to maximize the thrust force while maintaining a sufficient safety margin for a breakdown voltage, the positive-sequence voltage should be eliminated, because the positive-sequence voltage is ineffective in producing thrust force; it just reduces the margin. Designing a slider with well-balanced capacitances is challenging because symmetrical patterning of three-phase electrodes is difficult due to the required bridging of lines. Finding a practical solution in patterning is left for future works.

VI. Conclusion

Modeling and analysis of a linear REIM with unbalanced capacitance have been reported in this paper. The developed model relates capacitances of the electrodes to the symmetrical components of the slider voltage and to the thrust force of the motor. Nonideal effects indicated by the model, e.g., the imbalance of slider voltage, the presence of multiple frequencies in the voltage, and the resulting thrust force ripple, were confirmed by experiments using an unbalanced motor.

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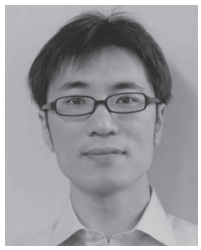
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